

~ 200 facility locations with 50 mile buffer distances

- 8 different types of facilities: mills, conversion, enrichment, fuel fab, power plants, onsite storage, proposed interim storage, and proposed repositories
 - o 4 sub-groups: front-end, power production, back-end, and proposed back-end facilities
 - o *Mines don't have buffer zones. I will instead count mines within county lines. Straightforward enough to do.*
- Buffers will be weighted proportionally to a given scale attribute
 - o rem, gigawatts, megaton uranium produced, megaton uranium stored, etc
 - o (I have this data; I imagine the calculated % of people within the buffer can just be multiplied by this scale)
- Buffers are allowed to overlap and double-count populations. This represents a higher risk imposed on any given person (e.g. Person A who lives near 2 reactor is exposed to more risk than Person B who lives near 1, therefore we double count them to show that they are exposed to greater risk)

Demographics

- 1950-2020 (or to 2023)
- County level
- 5 year increments
- Total population
- Race and ethnicity (White, Black, Asian, American Indian, Hispanic)
- Age
- Sex
- Some sort of income metric
 - o Income relative to poverty level, income per capita, or median income
 - o I was having trouble finding a consistent metric over time
- Educational attainment (no education, high school/GED, bachelors+)
- If regularly available
 - o Urban/rural/farm status
 - o Median housing cost
 - o Employment rate
- *Can a feature layer actually handle a 3D array? With dimensions of county location/FIPS code, demographic category, and year?*

Analysis

- Calculate % of county area w/in a buffer zone (overlapping buffers allowed)
- For number-based census data (number of women, number of Hispanic, number with high school diploma)
 - o Weight raw number with % of county area calculated
- For statistic-based data (poverty rate, employment rate)

- Calculate proportion of county within the buffer (buffer X is 35% county A, 25% County B, and 40% county C). Use these proportions to recalculate the statistic of interest within the buffer
- Weight with capacity scale of each facility
- *Results in a 3D array, with dimensions of demographic statistic, year, and fuel cycle group category*